A Chebyshev-type integral inequality for positive bounded increasing functions, with equality only for constants

The Univalent Foundations Program

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Editorial note. The counted statement is the author's 'lem:int_fg' (source lines 1905-1912, Lemma 4.17); the proof environment starts on the next line, so the pairing is by adjacency.

Editorial note. The retained counter values (section 4, theorem offset 16) make the counted lemma print as Lemma 4.17; this article carries no label-number dump, so the number was inferred by counting the theorem-class environments of section 4 and is reported with that caveat.

Editorial note. No retained line refers to a label, citation or figure outside the retained span, so the package declares no external reference.

Editorial note. The retained preamble is the publisher layout adapter followed by the native author macro and theorem block of the pinned source.

Editorial scope. One original result of the pinned publisher article, counted once: the integral inequality for increasing functions with its equality case (author label lem:int_fg).

Allowed standard prerequisites (editorial).

- Comparison of measures. If one measure is dominated by another then the integral of an increasing function against the first is at most its integral against the second.. It is the step that converts the domination nu <= mu into the inequality at source line 1916.
- Distribution functions. The mass identity and the domination are checked on the cumulative distribution functions of mu and nu.. They are computed explicitly at source line 1916.
- Indicator functions of intervals. Positive combinations and limits of indicator functions of intervals generate the positive measurable functions used here.. They are the test functions in the equality analysis at source line 1918.

We will need the following little technical result.

Lemma 4.17. *Take $a < b$ in $\mathbb{R}$ and f and g two positive bounded increasing functions from $]a, b[$ to $\mathbb{R}$. Then*

$$\frac{1}{b-a}\int_a^b fg\,\mathrm{d}\lambda \geq \left(\frac{1}{b-a}\int_a^b f\mathrm{d}\lambda\right)\left(\frac{1}{b-a}\int_a^b g\mathrm{d}\lambda\right),$$

with equality if and only if f or g is constant.

Proof. The measures $\mu = f\lambda$ and $\nu = \frac{1}{b-a}\int_a^b f\,\mathrm{d}\lambda$ have the same mass $\int_a^b f\,\mathrm{d}\lambda$. Besides, $\nu \preceq_{\mathrm{sto}} \mu$; indeed, for any $c \in]a, b[$, since f is increasing:

$$\frac{1}{c-a}\mu(]-\infty, c]) = \frac{1}{c-a}\int_a^c f\,\mathrm{d}\lambda \leq \frac{1}{b-a}\int_a^b f\,\mathrm{d}\lambda = \frac{1}{c-a}\nu(]-\infty, c]).$$

Hence, as g is increasing, $\int g\,\mathrm{d}\mu \geq \int g\,\mathrm{d}\nu$, equivalent to the wished inequality.

Plainly, equality occurs if f or g is constant. If f is not constant, we in fact proved that equality in $\int g_c \mathrm{d}\mu \leq \int g_c \mathrm{d}\nu$ holds for $g_c = \mathbb{1}_{[c,b]}$ if and only if $c \in \{a, b\}$. This remains true for positive combinations of functions g_c and, being a little careful, for limits of them. □